

WICHITA EISENHOWER, KS, USA

WMO: 724500

Lat: 37.648N

Lon: 97.430W

Elev: 403

StdP: 96.58

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03928

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.4	-10.9	-19.2	0.7	-11.1	-16.9	0.9	-9.2	14.0	5.8	12.6	3.6	4.7	0	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	38.1	23.2	36.3	23.4	34.5	23.3	25.5	32.8	24.9	32.3	24.3	31.7	6.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.3	29.0	22.9	18.5	28.2	22.2	17.8	27.6	80.7	32.8	77.8	32.3	75.2	31.7	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.7	11.5	10.5	-17.1	40.3	2.6	1.7	-19.0	41.6	-20.5	42.6	-22.0	43.6	-23.9	44.8		
			-17.6	26.9	2.5	0.8	-19.4	27.4	-20.9	27.9	-22.3	28.4	-24.1	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	0.8	3.0	8.5	13.7	19.4	25.0	27.5	26.7	22.3	15.2	8.1	1.9
	DBStd	10.63	5.84	6.29	6.10	5.10	4.41	3.44	3.17	3.12	4.34	5.09	5.28	5.48
	HDD10.0	979	288	206	98	22	1	0	0	0	1	13	96	255
	HDD18.3	2428	544	428	306	152	42	2	0	0	14	123	308	509
	CDD10.0	2585	2	11	53	134	292	449	542	517	369	174	38	4
	CDD18.3	993	0	0	3	14	75	201	283	258	133	26	1	0
	CDH23.3	10833	0	1	28	151	679	2148	3398	2881	1298	241	9	0
	CDH26.7	5208	0	0	4	33	232	1017	1818	1469	564	70	1	0
Wind	WSAvg	5.1	4.9	5.1	5.7	6.0	5.2	5.3	4.6	4.3	4.7	5.0	5.1	4.8
Precipitation	PrecAvg	804	20	27	59	69	111	118	90	92	85	67	37	31
	PrecMax	1367	81	99	233	253	334	266	246	304	329	239	149	120
	PrecMin	309	0	0	0	6	13	10	1	4	12	1	0	0
	PrecStd	193	19	24	46	47	69	59	59	65	73	51	34	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	22.5	26.4	30.0	34.2	38.3	40.6	40.1	37.2	31.7	24.5	18.7
		MCWB	11.0	12.6	15.5	18.4	21.4	23.2	23.1	22.6	22.0	19.8	15.5	12.3
	2%	DB	15.4	19.0	23.3	27.1	31.4	35.7	38.5	37.6	34.1	28.5	21.7	15.6
		MCWB	8.3	11.3	15.1	17.5	21.3	23.4	23.2	23.2	22.2	18.4	14.1	10.3
	5%	DB	12.5	15.9	20.9	24.9	29.2	33.9	36.6	35.7	32.2	26.1	18.9	12.9
		MCWB	7.1	9.8	13.8	16.6	20.7	23.2	23.7	23.3	22.0	17.8	12.8	8.5
	10%	DB	9.9	13.0	18.4	22.6	27.2	32.1	34.6	33.6	30.0	23.5	16.5	10.4
		MCWB	5.7	8.3	12.4	15.7	20.1	22.6	23.7	23.1	20.9	17.0	11.8	6.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	15.1	19.0	20.8	24.0	26.1	26.7	26.3	24.8	22.6	17.6	15.3
		MCDB	15.2	18.5	23.1	27.0	29.7	33.7	33.5	33.4	31.9	27.3	21.8	17.4
	2%	WB	9.8	12.8	16.9	19.6	22.9	25.1	25.6	25.3	23.7	21.1	15.6	11.6
		MCDB	13.7	16.9	20.7	24.3	28.6	32.4	32.9	32.6	30.7	25.4	19.1	13.6
	5%	WB	7.6	10.7	15.2	18.3	22.0	24.2	25.0	24.6	22.9	19.6	14.4	9.0
		MCDB	12.4	15.6	19.2	23.0	27.2	31.2	32.5	32.0	29.4	23.9	17.9	12.3
	10%	WB	5.7	8.0	13.0	16.8	21.0	23.5	24.3	23.9	22.1	17.9	12.5	6.8
		MCDB	9.7	12.9	17.7	21.5	25.9	30.1	32.0	31.1	28.3	22.2	16.2	10.3
Mean Daily Temperature Range	5% DB	MDBR	11.3	11.8	12.2	12.2	11.4	11.5	11.7	11.6	12.0	12.2	11.8	10.6
		MCDBR	16.2	16.6	16.1	15.2	13.7	13.6	14.5	14.4	13.9	15.1	15.1	14.7
		MCWBR	10.4	10.4	9.3	8.5	6.1	4.7	3.8	4.2	4.9	7.2	8.9	9.8
	5% WB	MCDBR	15.4	15.0	12.8	12.8	11.7	12.2	12.2	11.9	11.8	11.3	12.2	12.8
		MCWBR	10.5	10.4	8.9	8.3	6.2	5.1	4.5	4.6	4.8	6.8	8.6	9.7
Clear-Sky Solar Irradiance	taub	0.287	0.298	0.334	0.372	0.405	0.413	0.421	0.424	0.380	0.339	0.299	0.281	
	taud	2.499	2.479	2.403	2.322	2.306	2.313	2.329	2.276	2.377	2.479	2.520	2.524	
	Ebn at Noon	907	939	929	907	878	868	857	847	869	879	885	891	
	Edh at Noon	79	91	109	125	130	129	126	130	111	91	77	72	
All-Sky Solar Radiation	RadAvg	2.53	3.25	4.26	5.35	5.99	6.84	6.73	6.04	5.01	3.71	2.74	2.16	
	RadStd	0.25	0.47	0.29	0.35	0.53	0.50	0.41	0.34	0.36	0.42	0.28	0.21	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+127	N/A
	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page